Demo 3 Timed Review Test Results

1. Test Setup

- **Drafts:** 10 rough subcontractor-to-GC RFI forms (R01–R10), rewritten from real Dett Elementary RFIs (Addenda 5 and 6). R06, R09 and R10 are complete control drafts. Data: demo3/data/rough_drafts.csv.
- **Conditions: Without AI** = the rough draft as written. **With AI** = the same draft after the AI agent suggested changes (each with a source) and the subcontractor accepted or rejected each suggestion.
- **Agent:**Gemini. The agent received only the RFI form fields, never the answer key.
- **Subcontractor role:** one team member. The sub could only add information from the scenario brief and source documents.
- **Reviewers:** GC reviewers (GC1: Kevin Chen, GC2: Kevin's colleague), blind to the condition and to the answer key.
- **Assignment:** Form A (GC1) = R01–R05 without AI, R06–R10 with AI. Form B (GC2) = R01–R05 with AI, R06–R10 without AI. No reviewer saw both versions of the same draft. One practice RFI first, not counted.
- **Task:** For each RFI, the reviewer opened it (timer starts), decided "can move forward" or "needs more information" (timer stops), and wrote down every question they would ask the subcontractor.

2. Subcontractor Step (With AI)

ID	AI suggestions	Accepted	Rejected	Sub prep time	Notes
R01	6	6	0	6 min	Added rooms, sheets and keynotes; updated 6/A-703 to 1/A-703 (Addendum 5); added deadline, impact and photo description
R02	4	4	0	3 min	Added Hallway 101, A-210 detail 1, KN A.34, needed-by date

ID	AI suggestions	Accepted	Rejected	Sub prep time	Notes
R03	5	5	0	7 min	Added Staff Toilet 140 and 143; "P2" changed to P-201 and P-202 detail 2; "underfloor dwg" changed to P-100A
R04	4	3	1	4 min	AI suggested "A-301" for "A-30"; sub rejected it and entered A-301B elevations 1 and 3
R05	4	4	0	3 min	Added rooms 108B, 109A, 110B, 111A and sheets A-502, A-503
R06	0	0	0	1 min	Control: no changes needed
R07	4	4	0	3 min	Added E-202, E-203, E-204 and Addendum 5; added deadline and impact
R08	3	3	0	2 min	"E-60" changed to E-600 and PV-600; added both conduit sizes and a needed-by date

ID	AI suggestions	Accepted	Rejected	Sub prep time	Notes
R09	1	0	1	1 min	Control: AI asked for a location that was already given; sub rejected it (false alarm)
R10	0	0	0	1 min	Control: no changes needed

Totals: 31 suggestions, 29 accepted, 2 rejected. Median sub prep time: 3 min.

3. GC Review Results

ID	Without AI			With AI		
	Decision	Follow-ups	Review time	Decision	Follow-ups	Review time
R01	Needs more info	5	32 min	Move forward	0	9 min
R02	Needs more info	3	7 min	Move forward	0	4 min
R03	Needs more info	5 + 1 internal	40 min	Needs more info	1 + 1 internal	12 min
R04	Needs more info	4	8 min	Move forward	0	5 min
R05	Needs more info	3	6 min	Needs more info	1	4 min
R06	Move forward	0	5 min	Move forward	0	5 min
R07	Needs more info	4	7 min	Move forward	0	4 min
R08	Needs more info	3	6 min	Move forward	0	4 min
R09	Move forward	0	5 min	Move forward	0	5 min
R10	Move	0	5 min	Move	0	4 min

ID	Without AI			With AI		
	Decision	Follow-ups	Review time	Decision	Follow-ups	Review time
	forward			forward		

Remaining follow-ups With AI: R03 – "Please send the fixture schedule request as a separate RFI." R05 – "Please send the T-2 tile pattern question as a separate RFI." In both cases, the AI did not split a message that asked two questions.
Internal flag (both conditions): R03 says "not carrying extra fixtures unless directed," so the GC checks the contract with the PM. This is not a question for the sub and is counted separately.

4. Errors, False Alarms and Sources

Check	Result
Information errors in final With AI drafts	0 of 10
Wrong AI suggestions caught by the sub	1 (R04: suggested A-301 instead of A-301B)
False alarms on control drafts (R06, R09, R10)	1 of 3 (R09), rejected by the sub
Suggestions with a correct source	29 of 31

5. Summary

Measure	Without AI	With AI	Change
Median GC first-review time	6.5 min	4.5 min	−31%
Total follow-up questions to the sub	27	2	−93%
Median follow-ups per RFI	3	0	−3
RFIs marked "needs more information"	7 of 10	2 of 10	−5
Information errors in final drafts	n/a	0	no increase
Median subcontractor prep time	0 min	3 min	+3 min

6. What This Says About Our Thesis

Time. Median GC first-review time fell by 31%, which meets our 25% target. The biggest

savings were on the two complex drafts (R01 and R03). On simple drafts, the review was only 2–3 minutes faster, which matches our GC interview.
Back-and-forth. Follow-up questions fell from 27 to 2. Our GC interviewee said each round of back-and-forth costs a day or two, so this may save more time than the faster first review.
Errors. No information errors reached the GC, but only because the sub rejected one wrong AI suggestion. The AI also raised one false alarm on a complete draft. This supports our design: the AI suggests, and the sub makes the final decision.
Cost to the sub. The sub spent a median of 3 extra minutes per RFI, so part of the time moves from the GC to the sub.

7. Answer Key Issues Found

1. R04 and R07 list a missing impact as a planted issue, but the answer key has no matching follow-up question.
2. R03's cost hint is an internal GC check, so internal flags are now counted separately.
3. R08's Location field is empty, but a location problem is not listed in its planted issues.

8. Limitations

- Small sample: 10 drafts and 2 reviewers. Results show direction only.
- Our team wrote the drafts and the answer key, so the test may be easier than real RFIs.
- Reviewers only read the forms. They did not call the sub or visit the site, so times are shorter than in real work.
- Different reviewers saw each condition for the same draft, so some of the time difference may come from reviewer speed.

9. Next Steps for Demo 4

- Make the agent split messages that ask more than one question.
- Fix the answer key issues in section 7.
- Measure total time to an answer, including back-and-forth rounds, not only first-review time.